A closer analysis of (4.16) makes us expect that the last term gives rise to three different isobaric parabola depending on whether the nuclei are odd-A (even-odd or odd-even), odd-odd, or even-even (Fig. 4.6). In the first case, in which the mass number is odd, we find a single parabola (I); whether all beta decay leads to changes from odd-even to even-odd, etc. For even mass numbers one finds a double parabola (II) – (V). When the individual nuclear properties are considered, the difference between the curves for the odd-odd and even-even nuclei may lead to alternatives with regard to the numbers of possible stable isobars: it is possible to find three stable isobars (case V) although two (case IV) are more common. Although the odd-odd curve always must lie above the even-even curve, still an odd-odd nucleus may become stable, as is shown for case II.

4.8. THE MISSING ELEMENTS: $_{43}$TC AND $_{61}$PM

Among the stable elements between $_1$H and $_{82}$Pb two elements are "missing": atomic number 43, named *technetium* (Tc), and atomic number 61, *promethium* (Pm). Though these elements can be produced through nuclear reactions and also have been found to exist in certain stars, they are not found on earth because their longest lived isotopes have much too short half-lives for them to have survived since the formation of our planet. This can be understood by considering the valley of β-stability. For pedagogic reasons we will first discuss promethium.

4.8.1. Promethium

The valley of β-stability for $Z = 61$ shows a minimum around mass number $A = 146$, for which the isotopes are either of the even-even or of the odd-odd type. Thus the binding energy curve should exhibit two isobar parabolas, as illustrated in Figure 4.7; the decay energy Q is released binding energy. ^{146}Pm has a 5.5 y half-life and decays either by electron capture (63%) to ^{148}Nd or by β^--emission (37%) to ^{146}Sm, who both are more stable (i.e. have a larger nucleon binding energy); the nuclear binding energy is given on the vertical scale in the Figure. The curves shown in Figure 4.7 differ from those calculated from eqn. 4.14 by about 1 unit in Z due to deviations from the simple liquid drop model in the lanthanide region, see §6.4.

The two adjacent isobars, mass numbers $A = 145$ and $A = 147$, are of the odd-even and even-odd types, thus only one isobaric β-decay curve exists for each of these. The decay scheme for $A = 145$ follows curve I in Figure 4.6 for which $^{145}_{60}$Nd is the stable isobar. ^{145}Pm is the longest lived promethium isotope ($t_{½}$ 17.7 y). For $A = 147$, the stable isobar is $^{147}_{62}$Sm; the half-life of ^{147}Pm is 2.62 y, which makes it the most convenient radioisotope of promethium for use in experiments.

Promethium is a fission product (Ch. 5 and 19) and can be chemically isolated in pure form. It exhibits typical lanthanide properties and is used in technology and medicine as a radiation source (Ch. 18).

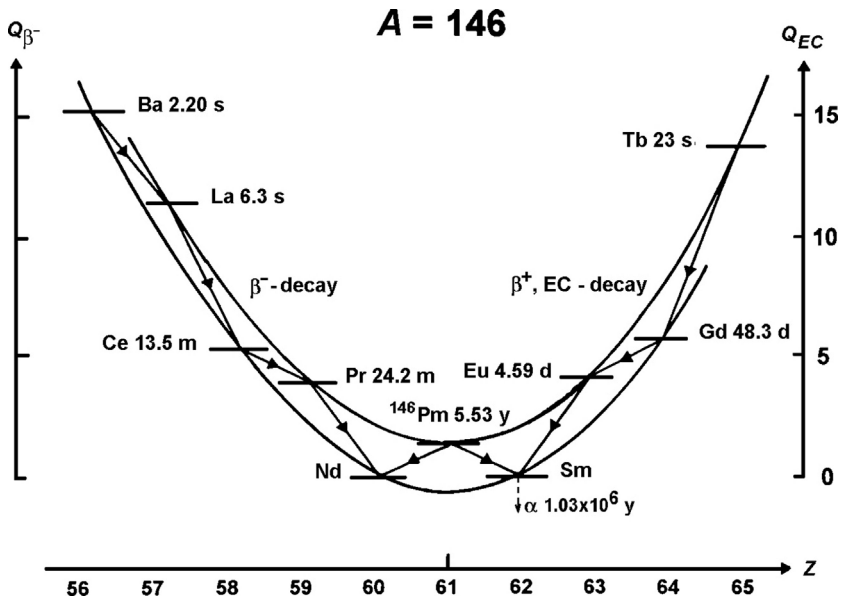


Figure 4.7 Decay scheme for $A = 146$, with isobar half-lives. Decay energy Q in MeV. ^{146}Nd and ^{146}Sm are β-stable.

4.8.2. Technetium

For technetium, $Z = 43$, the valley of β-stability has a minimum in the neighbourhood of $N = 55$ and thus, for $Z = 43$, A-values around 97 and 99 are most likely to be stable (recall that odd-odd nuclei are less stable than odd-even). If one considers all the isobars between $A = 95$ and 102 one finds that for each mass number in this range there is already at least one stable nuclide for the elements with $Z = 42$ (molybdenum) and $Z = 44$ (ruthenium). Since adjacent isobars cannot both be stable, this excludes the possibility of stable odd-even isotopes of technetium. The longest lived isotopes of technetium are those with $A = 97$ (2.6×10^6 y), $A = 98$ (4.2×10^6 y), and $A = 99$ (2.1×10^6 y). Figure 4.8 shows the decay scheme for $A = 99$, which is taken out of a standard Isotope Table; the vertical axis shows the relative binding energies (broken scale). The Figure illustrates the information normally presented in isotope tables, and will be further explained in subsequent chapters.

Hundreds of kilograms of ^{99}Tc and its mother ^{99}Mo are formed every year as fission products in nuclear reactors, and 10's of kg of Tc have been isolated and studied chemically. Its properties resembles those of its homologs in the Periodic Table - manganese and rhenium. Figure 4.8 shows decay schemes for mass number 99: the upper one from Shirley et al, 1986, the lower one from Dzhelepov et al, 1961; more detailed schemes appear in both references. The ones shown in Figure 4.8 were chosen

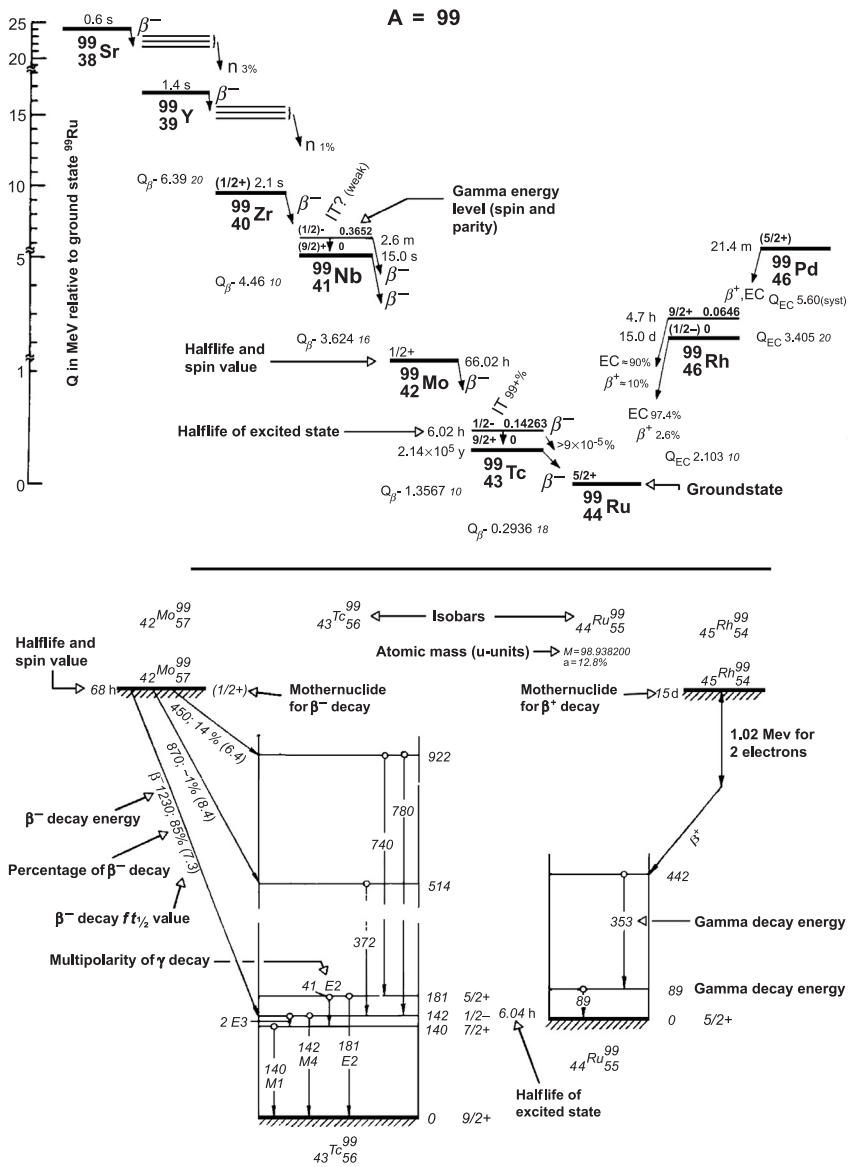


Figure 4.8 The $A = 99$ decay scheme. *(From E. Browne, R. B. Firestone and V. S. Shirley, Table of Radioactive Isotopes, and from B. S. Dzhelepov and L. K. Peker, Decay schemes of radioactive nuclei.)*

for pedagogic reasons, and, for this purpose, we have also inserted explanations, some of which will be dealt with later. Older references are often still useful for rapid survey, while the newest ones give the most recent information and refined numerical data.

The upper left part of Figure 4.8 shows a decay chain from fission of ^{235}U that ends in ^{99}Ru, the most stable isobar of $A = 99$. The lower diagram shows that the ^{99}Mo β^- decays all reaches the spin/parity ½-level (see § 3.6.1 and Ch. 11), designated ^{99m}Tc; this isomer decays with $t_{½}$ 6.02 h to long-lived ^{99}Tc, emitting a single γ of 0.142 MeV (> 99%, see upper diagram). The isomer ^{99m}Tc is a widely used radionuclide in nuclear diagnostics (§18.5), and can be conveniently "milked" from its mother ^{99}Mo, see §5.16.

4.9. OTHER MODES OF INSTABILITY

In this chapter we have stressed nuclear instability to beta decay. However, in §4.4 it was learned that very heavy nuclei are unstable to fission. There is also a possibility of instability to emission of α-particles in heavy elements and to neutron and proton emission.

Nuclei are unstable to various forms of decay as indicated in Figure 4.1. For example, making a vertical cut at $N = 100$, the instability from the top is first proton emission, then, α-emission (for $N = 60$ it would instead be positron emission or electron capture, as these two processes are about equally probable), and, after passing the stable nuclides (the isotones ^{170}Yb, ^{169}Tm and ^{168}Er), β^-emission and, finally, neutron emission. This is more clearly indicated in Appendix C, and for the heaviest nuclides (i.e. $Z \geq 81$) in Figures 5.1 and 14.1. For α-decay the Figure indicates that for $A > 150$ ($Z \geq 70$, $N > 80$) the nuclei are α-unstable, but in fact α-decay is commonly observed only above $A \approx 200$. This is due to the necessity for the α-particle to pass over or penetrate the Coulomb barrier (cf. §6.7.3). Although neutron and proton emissions are possible energetically, they are not commonly observed as the competing β-decay processes are much faster.

4.10. EXERCISES

4.1. Calculate the nucleon binding energy in ^{24}Mg from the atomic mass excess value in Table 4.1.

4.2. How many times larger is the nucleon binding energy in ^{24}Na than the electron binding energy when the ionization potential of the sodium atom is 5.14 V?

4.3. Assuming that in the fission of a uranium atom an energy amount of 200 MeV is released, how far would 1 g of ^{235}U drive a car which consumes 1 liter of gasoline (density 0.70 g cm^{-3}) for each 10 km? The combustion heat of octane is 5500 kJ mole^{-1}, and the combustion engine has an efficiency of 18%.

4.4. Estimate if fusion of deuterium into helium releases more or less energy per gram of material consumed than the fission of uranium.

4.5. When a neutron is captured in a nucleus, the mass number of the isotope increases one unit. In the following Table mass excess values are given for three important isotope pairs: